MALAVIYA NATIONAL INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

AI-Based Attendance Management System

Project Report

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Certificate

This is to certify that this project titled **AI-Based Attendance Management System** has been completed by **Ketan Sharma** under my supervision.

Guide Signature

Abstract

The AI-Based Attendance Management System uses facial recognition and machine learning techniques to automate classroom attendance. The system captures student faces, recognizes identities, and records attendance automatically. This reduces manual effort and improves accuracy.

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Chapter 1

Introduction

Attendance monitoring is an essential activity in educational institutions. Traditional attendance systems are time-consuming and prone to human error. Artificial Intelligence provides an automated solution using computer vision.

1.1 Problem Statement

Manual attendance:

- Takes time.
- Can contain errors.
- Allows proxy attendance.

Chapter 2

Objectives

The objectives of this project are:

1. Automate attendance using facial recognition.
2. Reduce proxy attendance.
3. Generate attendance reports automatically.
4. Store records securely in a database.

Chapter 3

Literature Review

Several AI-based attendance systems have been proposed using:

- OpenCV
- Haar Cascade
- FaceNet
- Deep Learning CNN models

Chapter 4

Methodology

4.1 System Architecture



4.2 Workflow

1. Capture Image.
2. Detect Face.
3. Extract Features.
4. Compare with Database.
5. Mark Attendance.

4.3 Recognition Accuracy

Recognition accuracy is calculated as

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions} \times 100$$

Chapter 5

Results

5.1 Attendance Summary

Table 5.1: Attendance Records

Student	Present	Attendance (%)
Ketan Sharma	28	93.3
Rahul Kumar	27	90.0
Aman Singh	30	100

Chapter 6

Discussion

The proposed system successfully recognized registered students with high accuracy. Performance depends upon lighting conditions and camera quality.

Chapter 7

Conclusion

The AI-Based Attendance Management System provides an efficient and reliable attendance solution using computer vision. Future work includes mobile integration and cloud synchronization.

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[3] Florian Schroff et al., FaceNet: A Unified Embedding for Face Recognition.